CS3 RESEARCH EXPERIENCE FOR TEACHERS (RET)

CVAT Tutorial

Labeling images with bounding boxes and exporting them for YOLO

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OVERVIEW

Twenty steps from raw images to a trained detector

CREATE THE TASK

1. Open the CVAT website
2. Create a new task
3. Press *Add label*
4. Enter labels and press *Continue*
5. Check your label classes
6. Drag and drop your images
7. Press *Submit & Open*

ANNOTATE

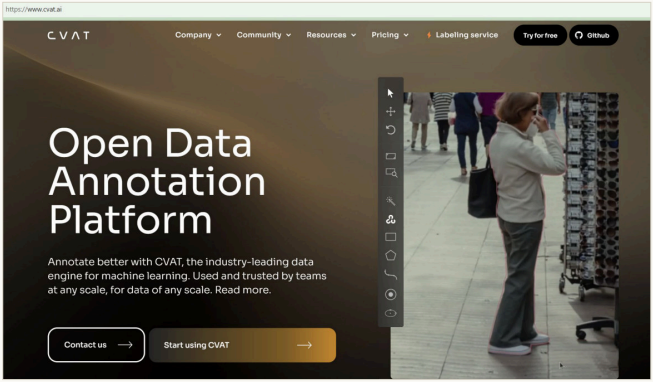
8. Open the job
9. Choose the rectangle tool
10. Draw a box around each object
11. Delete mistakes
12. Save your changes
13. Go back to *Tasks*
14. Open the task

EXPORT AND TRAIN

15. Export the annotations
16. Choose *YOLO 1.1*
17. Open the download link
18. Download the export
19. Arrange the dataset folders
20. Train with four lines of code

STEP 1

Open the CVAT website

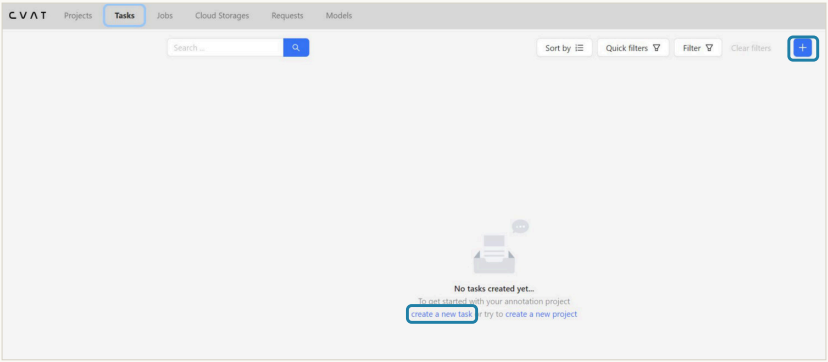


Go to the CVAT website in your browser:

https://www.cvat.ai/

STEP 2

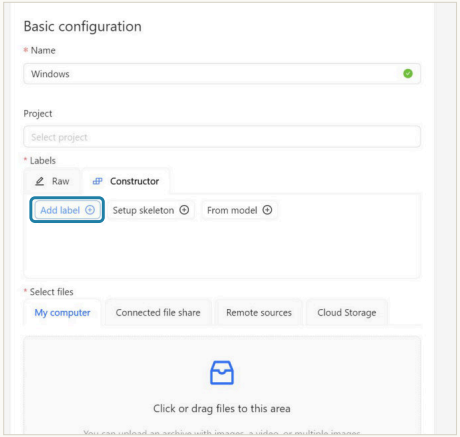
Create a new task



On the *Tasks* page, press the **+** button at the top right, or the *create a new task* link.

STEP 3

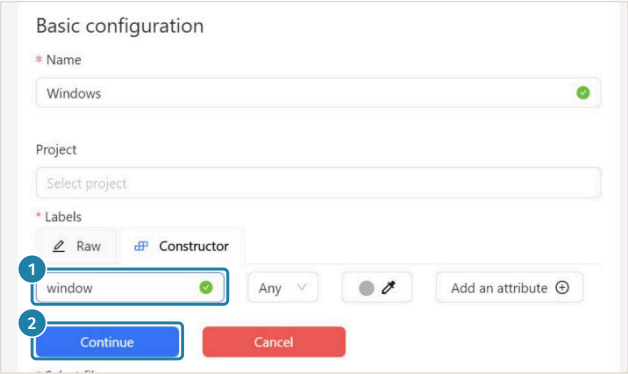
Press *Add label*



Give the task a name (here, *Windows*), then press *Add label* under *Labels*.

STEP 4

Enter each label and press *Continue*

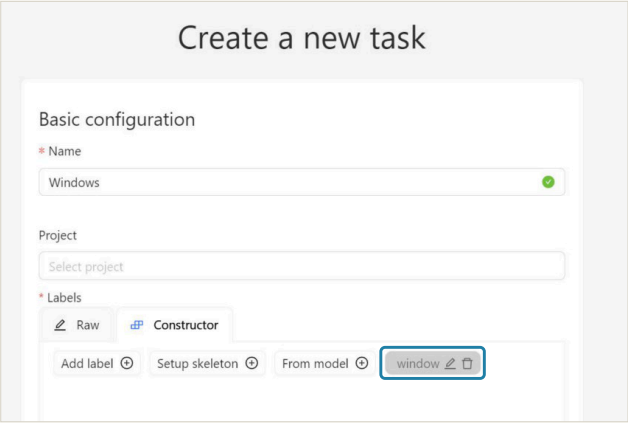


1. Type the name of a label class, such as *window*.
2. Press *Continue* to add it, then type the next one.

When all your classes are in, press *Cancel* to close the label form.

STEP 5

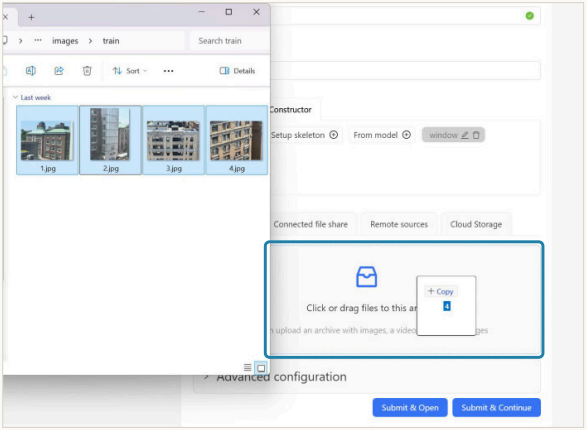
Check that your label classes appear



If everything worked, each class you added shows up as a tag next to *From model*.

STEP 6

Drag and drop your images

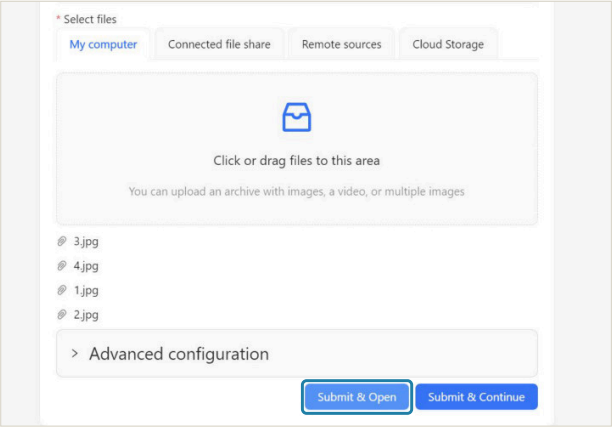


Drag your image files from a folder onto the *Click or drag files to this area* box.

You can also drop a single `.zip` file that contains all of them.

STEP 7

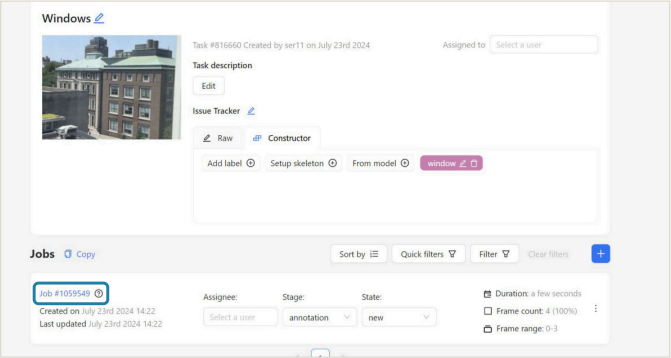
Press *Submit & Open*



The uploaded files are listed under the drop area. Press *Submit & Open* to create the task.

STEP 8

Open the job



Under *Jobs*, press the link named *Job #XXXXX*.

If you do not see this screen, click *Tasks* at the top of the page and open your task.

STEP 9

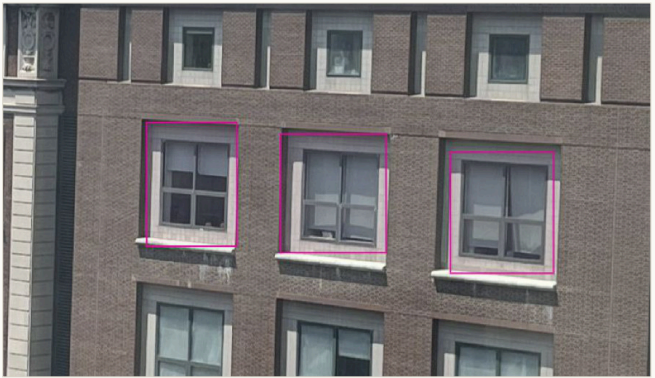
Choose the rectangle tool and press *Shape*



1. Select the rectangle tool in the left toolbar.
2. Check the label, then press *Shape*.

STEP 10

Draw a box around each object

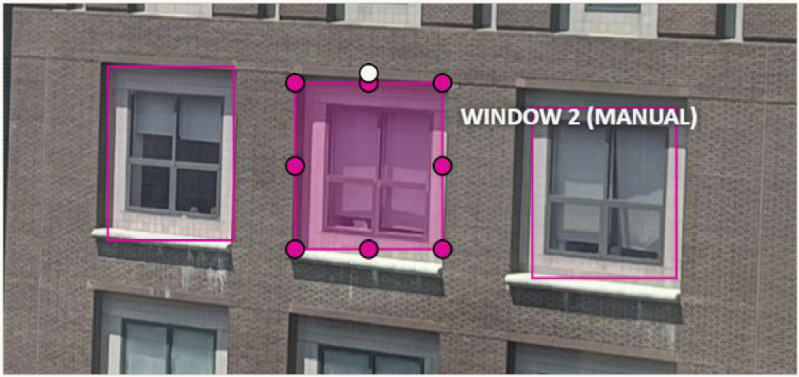


Click and drag with the mouse to draw a box tightly around one object.

Press N to start a new annotation, and repeat for every object in the image.

STEP 11

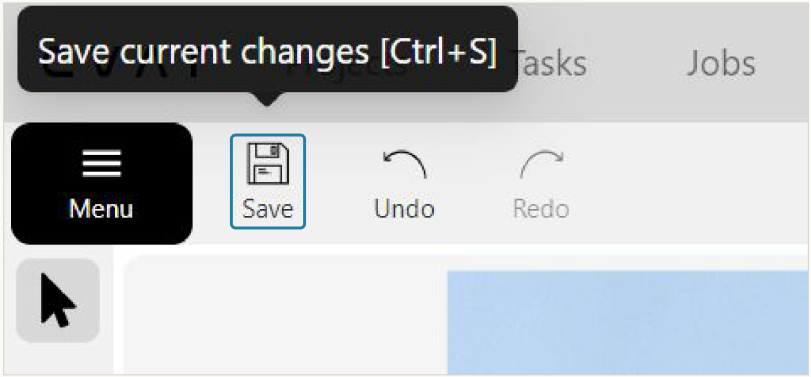
Delete mistakes with the *Delete* key



Hover over an annotation so that it is highlighted, then press `Delete` to remove it.

STEP 12

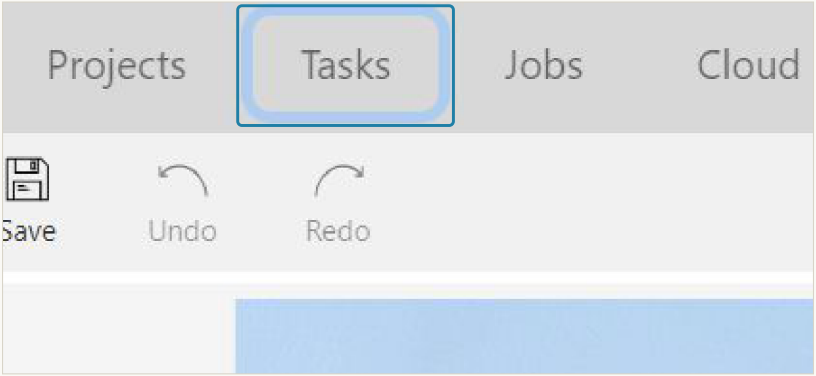
Save your changes



Click the *Save* button in the top toolbar, or press `Ctrl+S`.

STEP 13

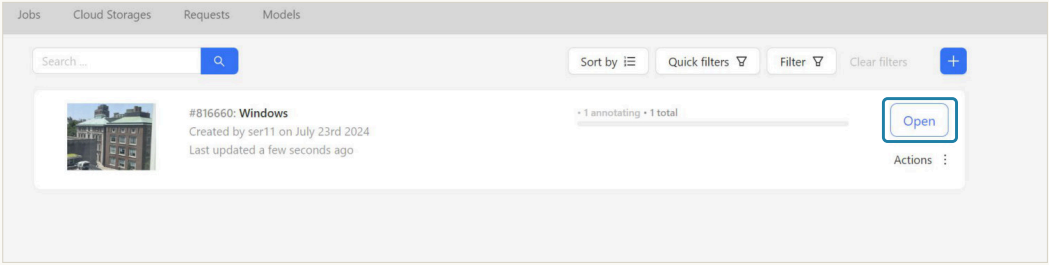
Go back to *Tasks*



When you have labeled every image, click *Tasks* in the top menu.

STEP 14

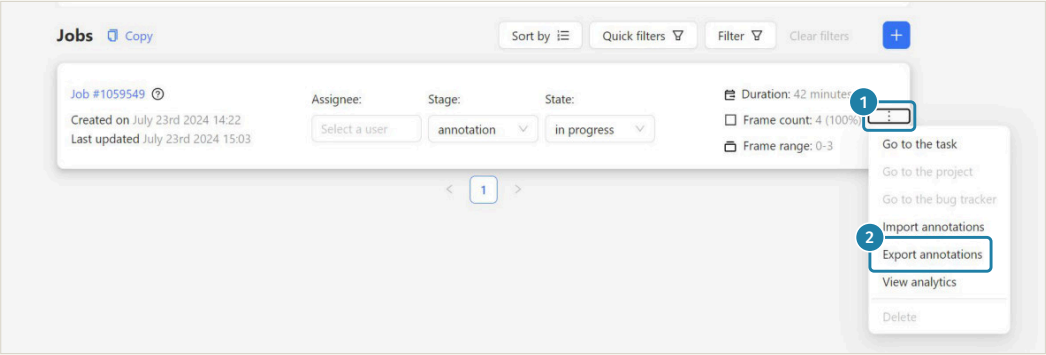
Open the task



Find your task in the list and press *Open*.

STEP 15

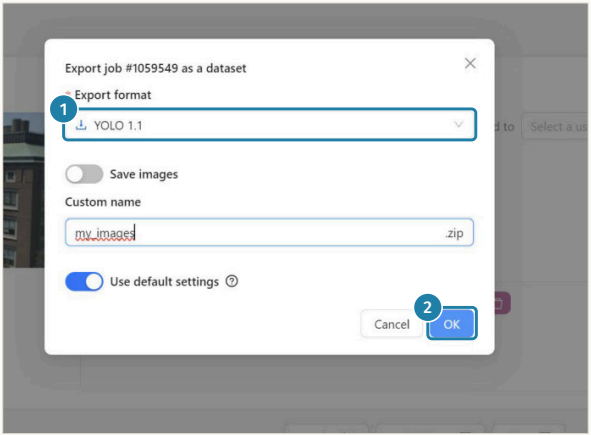
Export the annotations



1. In the job row, click the three dots.
2. Press *Export annotations*.

STEP 16

Choose *YOLO 1.1* and press *OK*



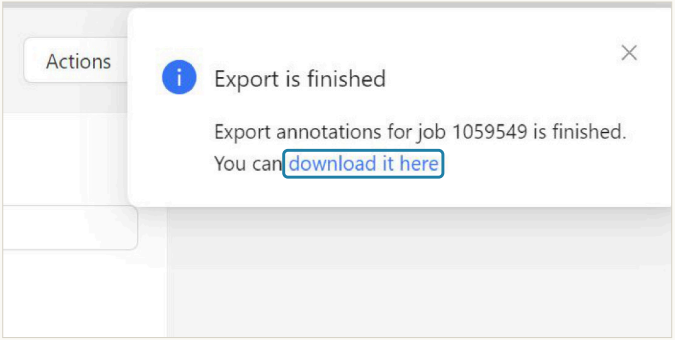
1. Set *Export format* to *YOLO 1.1*.
2. Press *OK* to start the export.

Very important

Select *YOLO 1.1* here. This is the annotation format the training step expects.

STEP 17

Open the download link



When the message *Export is finished* appears, click *download it here*.

STEP 18

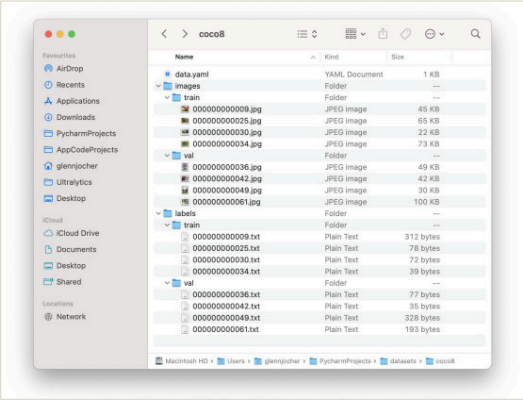
Download the export



1. On the finished export, click the three dots.
2. Press *Download* to save the `.zip` file.

STEP 19

Arrange the dataset folders



Create a folder structure like the one shown, with images and labels split into `train` and `val`:

```
coco8/
  data.yaml
  images/
    train/  000000000009.jpg, ...
    val/
  labels/
    train/  000000000009.txt, ...
    val/
```

Each image has a label file with the same name. An example dataset is provided as `windows.zip`.

STEP 20

Train your model with four lines of code

```
import os
import ultralytics
model = ultralytics.YOLO('yolov8n.pt')
results = model.train(data = 'data.yaml', epochs=100)
```

What happens

The code loads the pretrained `yolov8n.pt` model and trains it for 100 epochs on the dataset described in `data.yaml`. The Colab YOLOv8 training tutorial walks through running it step by step.